```python
import matplotlib.pyplot as plt

def simple_linear_regression(x, y):
    n = len(x)
    sum_x = sum(x)
    sum_y = sum(y)
    sum_xy = sum(x[i] * y[i] for i in range(n))
    sum_x2 = sum(x[i] ** 2 for i in range(n))

    b = (n * sum_xy - sum_x * sum_y) / (n * sum_x2 - sum_x ** 2)
    a = (sum_y - b * sum_x) / n

    return a, b

# Sample data
x = [1, 2, 3, 4, 5]
y = [2, 4, 5, 4, 5]

# Calculate coefficients
a, b = simple_linear_regression(x, y)

print("Intercept (a):", a)
print("Slope (b):", b)

# Predict y for a given x
x_new = 6
y_pred = a + b * x_new
print("Predicted y for x =", x_new, "is", y_pred)

# Graph Plotting

y_reg = [a + b * xi for xi in x]

plt.scatter(x, y, label="Actual Data")
plt.plot(x, y_reg, label="Regression Line")
plt.scatter(x_new, y_pred, label="Predicted Point")

plt.xlabel("X")
plt.ylabel("Y")
plt.title("Simple Linear Regression using Least Squares Method")
plt.legend()
plt.grid(True)
plt.show()
```

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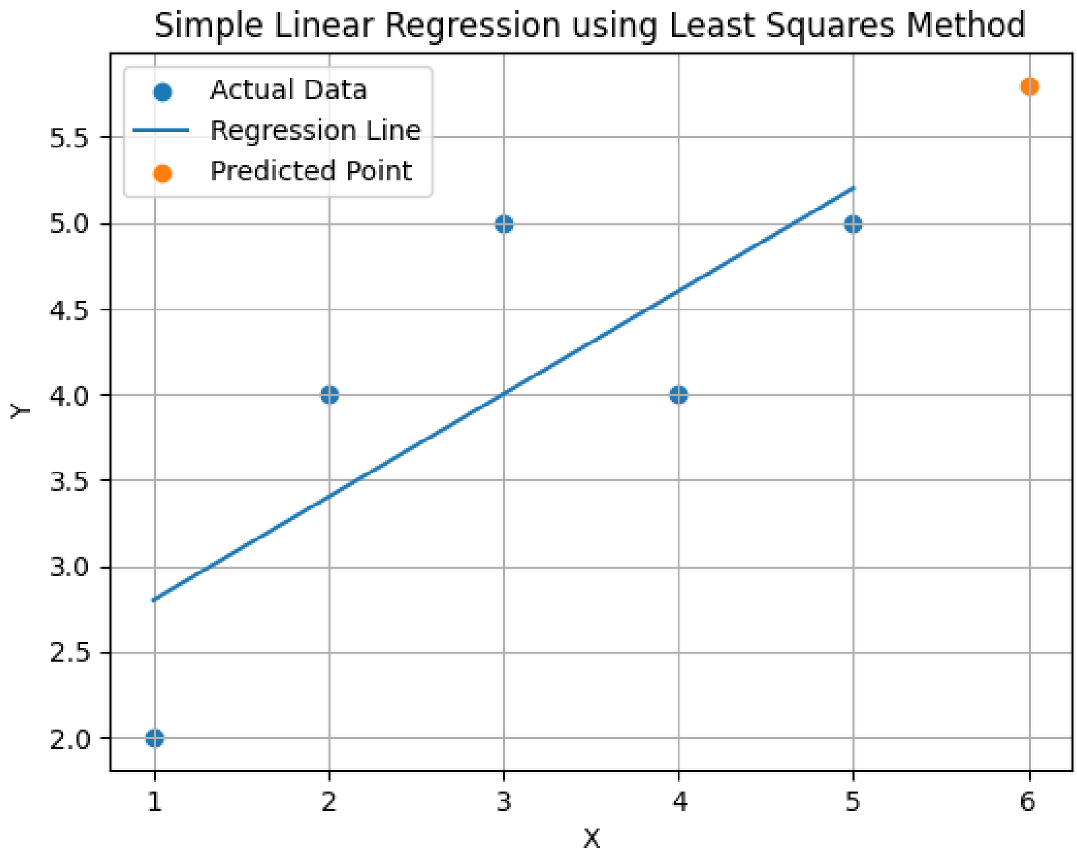
Intercept (a): 2.2
Slope (b): 0.6
Predicted y for x = 6 is 5.8
```


Simple Linear Regression using Least Squares Method
Actual Data
Regression Line
Predicted Point
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
Y
1
2
3
4
5
6
X